

Literature Review and Novelty Assessment

Proposed framework: gaze + mouse/interaction behavior + independent environment camera for exam-malpractice detection

Executive summary

The proposed framework is **not novel in its individual sensing components**: gaze or head-pose analysis, browser/window activity monitoring, mouse or keystroke behavior, person/object detection, and multi-camera or CCTV surveillance have all appeared separately in the literature. More importantly, several prior systems already combine two of the proposed dimensions. Atoum et al. combine gaze estimation, active-window monitoring, phone/text detection, audio, and two distinct cameras, although the second camera is a wearable field-of-view camera rather than a fixed room/CCTV camera and the paper does not present mouse-trajectory analytics ¹. Hu et al. use **three separate camera perspectives**—face, overhead, and horizontal—together with gaze recognition and detection of phones, paper, books, and people, but do not include mouse or keystroke behavior ⁵. Li and Li combine a webcam, a follower camera, gaze-view tracking, and custom object detection, again without a mouse channel ⁴.

Within the indexed and publicly accessible literature searched for this assessment, **no verified paper was found that clearly implements all three of the following as one integrated online-exam system: (i) gaze/eye tracking, (ii) mouse or equivalent interaction-behavior analytics, and (iii) a distinct independently angled environment camera**. The closest mouse-plus-video study is the CHI paper by Li et al., which analyzes exam video and mouse movement data, but it does not establish a gaze estimator or a separate environment camera ⁶. The closest three-camera/gaze/object study is Hu et al. ⁵. The strongest historical multimodal precedent is Atoum et al. ¹, which makes a broad claim of multimodal proctoring—including an active-window channel—but still does not equal the proposed continuous mouse-dynamics channel and does not propose CCTV extensibility.

Accordingly, the defensible conclusion is “**novel in the proposed combination and deployment architecture, but not novel in the individual components.**” The claim should not be written as “the first multimodal proctoring system” or “the first system using multiple cameras,” because those claims are contradicted by prior work ¹ ³ ⁴ ⁵ ⁷ ⁸. A narrower and stronger contribution would be a rigorously evaluated **three-stream fusion model** in which gaze, computer-interaction behavior, and an independently

positioned room camera are time-synchronized and fused into calibrated, explainable incident scores. The CCTV/physical-hall extension is also not established as a clear prior proposal in the sources reviewed, but it should be framed as a proposed cross-context deployment and validated rather than asserted as an unqualified first.

Important interpretation: “Active-window/tab-switch detection” is not automatically equivalent to “mouse dynamics.” A system may detect that a browser window lost focus without modeling cursor trajectories, click timing, idle behavior, pointer entropy, or anomalous interaction sequences. This distinction is central to the novelty analysis.

Full paper list for Task 1 — combined or multimodal systems

The following table emphasizes papers that combine at least two relevant modalities or sensing perspectives. “Separate environment camera” means a camera with a distinct viewpoint intended to see the workspace, forward field of view, room, desk, or surrounding area; a second face-facing webcam or a second crop of the same video does not qualify.

Paper	Citation and link	Modalities actually described	Data, hardware reported result
Automated Online Exam Proctoring	Y. Atoum, L. Chen, A. X. Liu, S. D. H. Hsu, and X. Liu, <i>IEEE Transactions on Multimedia</i> , 19(7), 1609–1624, 2017. DOI: 10.1109/TMM.2017.2656064 1	Two cameras plus microphone; user verification; text, speech, active-window, gaze, and phone detection; temporal joint decision.	Webcam above/integrate monitor faces tl examinee; “we is worn or attac eyeglasses and the examinee’ view. OEP data sessions/subject cheating types, synchronized a two videos. The report SVM/RBF experiments an the OEP datase
Multi-Modal Online Exam Cheating Detection	A. M. Abozaid and A. Atia, in <i>2022 International Conference on Electrical, Computer and Energy Technologies (ICECET)</i> , 2022. DOI: 10.1109/ICECET55527.2022.9873527 2	Indexed abstract confirms head pose, object detection, and eye-gaze estimation using a webcam. Search records also mention mouse-event monitoring, but the accessible abstract does	Evaluated on 25 over four exam indexed abstrac 95.69% event-d accuracy. The II was access-blo during verificat complete meth

		not clearly establish mouse trajectories as an independent fused modality.	hardware descr should be chec against the pub before making a claim.
Towards Effective and Efficient Online Exam Systems Using Deep Learning-Based Cheating Detection Approach	S. Kaddoura and A. Gumaiei, <i>Intelligent Systems with Applications</i> , 16, 200153, 2022. DOI: 10.1016/j.iswa.2022.200153 3	Front-camera and back-camera video plus microphone audio; CNN-based visual classifiers and Gaussian/DFT speech detection; soft-voting decision fusion.	The system pro sequences from back cameras a chunks. Science reports a multi-fusion architect the accessible abstract/methc does not identify analytics or a d gaze estimator.
Automated Online Proctoring System Using Gaze View Tracking and Custom Object Detection	B. Li and E. Li, in <i>2022 IEEE MIT Undergraduate Research Technology Conference (URTC)</i> , pp. 1–5, 2022. DOI: 10.1109/URTC56832.2022.10002245 4	Two cameras—a webcam and follower-cam—work with facial landmarks and a mathematical 3-D gaze-motion formulation; custom object detection identifies configured cheating targets.	The indexed ab confirms a prot trials for real-tir tracking and au target detection camera pair is c but the availabl does not concl establish the fo camera as a fixe room/CCTV view Numeric perfor not stated in th accessible abstr
Multi-Perspective Adaptive Paperless Examination Cheating Detection System Based on Image Recognition	Z. Hu, Y. Jing, G. Wu, and H. Wang, <i>Applied Sciences</i> , 14(10), 4048, 2024. DOI: 10.3390/app14104048 5	Three distinct camera perspectives: overhead for desktop tools, face for gaze, and horizontal for the forward field of view. Swin Transformer gaze recognition; Lightweight-YOLOv5-Coordinate Attention for books, paper, phones, and people; BP/MLP decision model.	Online simulati 35 FPS and 95% recognition rate system adaptiv overhead or ho object detectio gaze direction.
A Visual Analytics Approach to Facilitate	H. Li, M. Xu, Y. Wang, H. Wei, and H. Qu, <i>CHI ’ 21</i> ,	Exam video records plus mouse movement data;	Evaluated throu scenarios, a use

the Proctoring of Online Exams	article 682, pp. 1–17, 2021. DOI: 10.1145/3411764.3445294 6	suspected head and mouse movements are detected and visualized at multiple levels of detail.	and expert inte The source doe establish gaze e multiple camer separate enviro camera.
Multimodal Transformer Framework for Real-Time Cheating Detection in Online Assessments and E-learning Platforms	E. A. Alkinani, <i>SN Computer Science</i>, 7, article 101, 2026. DOI: 10.1007/s42979-025-04718-3 7	ViProctor-Transformer uses Video Swin-Transformer, Audio Spectrogram Transformer, cross-modal attention, temporal transformer, and auxiliary gaze/speech/phone cues.	Uses the MSU C dataset: synchr webcam and w camera video p microphone au 24 sessions and cheating behav Reports macro-AUROC 0.97, 15 lower false posi a ResNet50 bas at least 25 FPS c
The Three-Eyed Invigilator: An AI-Powered Interactive Rotator for Enhanced Online Exam Proctoring	I. Taj, A. S. Wazan, and A. Bey, <i>Neural Computing and Applications</i>, 38, article 106, 2026. DOI: 10.1007/s00521-026-11844-3 8	Interactive rotator for 360-degree environmental coverage; webcams, front/rear mobile cameras, microphones; face recognition, object detection, voice-activity detection.	Six scripted che scenarios; certa combinations r detection and p frame in 0.02 s. available from t corresponding request.
Online Student Authentication and Proctoring System Based on Multimodal Biometrics Technology	M. Labayen, R. Veja, J. Flórez, N. Aginako, and B. Sierra, <i>IEEE Access</i>, 9, 72398–72413, 2021. DOI: 10.1109/ACCESS.2021.3079375 9	Continuous face, voice, and typing biometrics; webcam/microphone monitoring; screenshots and device information; active-window/open-process/copy-paste monitoring; optional lockdown and human review.	Cloud-based sy online learning certification. Th describes conti authentication monitoring, but abstract does n one overall che classification ac

Task 1 synthesis

The literature shows that multimodal proctoring is not an empty field. Atoum et al. already provide a strong architecture combining multiple visual, audio, gaze, active-window, phone, and two-camera cues [1](#) . The later Hu system demonstrates that three distinct camera perspectives and gaze-plus-object fusion are feasible [5](#) . The Li et al. CHI system demonstrates that mouse movement can be analyzed jointly with exam video [6](#) . These papers, however, appear to occupy different regions of the modality space rather than implementing the exact intersection proposed here.

The evidence therefore supports a **combination-level gap**, not a component-level gap. A future paper should make the gap explicit as a reproducible matrix: continuous gaze features; continuous interaction features; independently positioned room camera; a defined temporal fusion model; and cross-domain deployment from remote exams to examination halls.

Full paper lists for Task 2 — individual components

(a) Eye/gaze tracking for proctoring or attention/cheating detection

Paper	Method and data	Reported results or limitations	Relevance
Automated Online Exam Proctoring — Atoum et al., 2017 1	Gaze estimation from webcam and wearcam; head-gaze vectors are combined with text, speech, active-window, phone, and identity cues. MSU OEP dataset contains 24 sessions and five cheating types.	The full paper reports component and classifier experiments, with RBF-SVM validation accuracies in the low-to-mid 90% range for some feature/PCA configurations.	Foundational multimodal gaze precedent; no continuous mouse-dynamics channel and no CCTV extension.
Cheating Detection in Browser-Based Online Exams Through Eye Gaze Tracking — Dilini et al., 2021 12	Browser plug-in tracks gaze through the built-in webcam; a generated eye-gaze dataset is analyzed using one-class SVM (OCSVM) anomaly detection.	The repository abstract does not state a numeric performance result.	Direct gaze-only browser proctoring; no mouse/keystroke or secondary camera.

Proctoring Online Exam Using Eye Tracking — Yaqub, Mohanty, and Suleiman, 2023 11	Privacy-preserving face hiding leaves the eyes visible; MTCNN, FaceNet/Siamese identity verification, EasyOCR, CNN-based iTracker, personalized linear calibration, and MediaPipe nose coordinates.	Five in-house cheating/exam videos; calibration error reduced to approximately 42% of raw error, about 3.97 cm on a 14-inch laptop for the best model; anomaly recall increased from 13% to 88% in the reported comparison.	Gaze-only anomaly detection with privacy protection; no mouse channel and no independent environment camera.
Modeling and Classification of the Behavioral Patterns of Students Participating in Online Examination — Ferdosi et al., 2023 10	MediaPipe facial landmarks; K-NN orientation classification; coordinated head, eye, and lip movements are converted into a thresholded cheating score.	Sixteen student volunteers and two mock exams; 100% accuracy for non-cheating cases and 87.5% for cheating cases at threshold 40.	Face/gaze behavior fusion, but no interaction logging or separate environment camera.
An Automated Online Proctoring System Using Attentive-Net to Assess Student Mischievous Behavior — Potluri, Venkatramaphanikumar, and Kishore, 2023 13	Attentive-Net face detection/alignment; FaceNet features; shallow CNN liveness; SolvePnp head-pose estimation; multiple-person detection. Uses CIPL and customized datasets.	Reports improved combined accuracy of 0.87. The system is visual and live-video based.	Head-pose/attention proxy and person context, but no dedicated gaze estimator, mouse channel, or independent environment camera.
AI-Assisted Gaze Detection for Proctoring Online Exams — Shih et al., 2024 26	Asynchronous gaze prediction per frame; scatter-plot interface allows proctors to navigate to similar gaze directions and suspicious timestamps. User study with 11 DET proctors and 300 sessions.	No final ML/human-complementarity accuracy is reported; the paper explicitly identifies calibration, direction-versus-screen-location, asynchronous operation, and human-judgment dependence as limitations.	Recent human-in-the-loop gaze triage; no mouse or secondary environment camera.

Multi-Perspective Adaptive Paperless Examination Cheating Detection System — Hu et al., 2024 ⁵	Face-camera gaze classification with Swin Transformer, linked to overhead/horizontal object detection and BP/MLP determination.	35 FPS and 95% average recognition in online simulations.	Strongest gaze-plus-independent-camera precedent, but no mouse/keystroke stream.
Automated Online Proctoring System Using Gaze View Tracking and Custom Object Detection — Li and Li, 2022 ⁴	Facial landmarks and 3-D gaze-motion formulation using webcam plus follower camera; TensorFlow custom object detector identifies configured targets.	Prototype trials report effective real-time gaze tracking and target detection; exact numeric performance is not present in the accessible abstract.	Close on gaze plus second camera, without mouse analytics or CCTV extension.

(b) Mouse movement, browser activity, keystroke dynamics, or behavioral biometrics

Paper	Method and data	Reported results or limitations	Relevance
PageFocus: Using Paradata to Detect and Prevent Cheating on Online Achievement Tests — Diedenhofen and Musch, 2017 ¹⁷	JavaScript records focus/blur behavior and detects switching to another window or browser tab. Study 1 used 115 laboratory and 186 online participants; Study 2 used 510 online participants.	PageFocus detected page abandonment with very high sensitivity and specificity and reduced cheating by a warning popup; performance-related incentives increased look-up cheating.	Direct precedent for tab/window focus, but not mouse trajectories or camera fusion.
A Visual Analytics Approach to Facilitate the Proctoring of Online Exams — Li et al., 2021 ⁶	Exam video records plus mouse movement data; suspected head and mouse movements visualized at three detail levels.	Usage scenarios, user study, and expert interviews support effectiveness and usability; no single classification accuracy is stated in the abstract.	Direct video-plus-mouse precedent; no gaze model or separate environment camera.
An Intelligent Approach for Fair Assessment of Online	Mouse dynamics only; compares KNN, SVC, random forest, logistic	LightGBM reportedly reaches up to 90% accuracy, 88%	Direct mouse-dynamics cheating classification;

Laboratory Examinations Based on Student's Mouse Interaction Behavior — Hosny et al., 2022 15	regression, XGBoost, and LightGBM on online laboratory exams, including MCQ/true-false and virtual remote-lab tasks.	precision, and 95% degree of separation.	explicitly avoids cameras and eye trackers.
Online Student Authentication and Proctoring System Based on Multimodal Biometrics Technology — Labayen et al., 2021 9	Face, voice, and typing biometrics for continuous authentication; device monitoring includes active windows, processes, screenshots, copy/paste, and lockdown.	Large-scale system and privacy/usability evaluations are described, but the abstract does not supply a single overall cheating accuracy.	Strong behavioral and active-device-monitoring precedent, but typing is primarily authentication and no gaze or room camera is established.
Keystroke Biometrics for Student Authentication: A Case Study — Morales and Fierrez, 2015 16	Keystroke dynamics during a real online introductory computer-science exam; key timing and typing-pattern behavior are used for student authentication.	The ACM abstract does not give a numeric result.	Foundational keystroke reference; authentication rather than cheating classification, and no gaze/camera data.
Multi-Modal Online Exam Cheating Detection — Abozaid and Atia, 2022 2	Indexed records clearly confirm head pose, object detection, and eye gaze; some search records mention mouse events, but the accessible abstract does not resolve whether mouse events are an independent fused stream.	Reports 95.69% event-detection accuracy for 29 students and four sessions. Full-text verification is needed for precise mouse claims.	Potentially relevant, but should not be presented as all-three prior art without reading the full publisher paper.
Online Student Authentication and Proctoring System Based on Multimodal Biometrics Technology —	Continuous typing recognition combined with face/voice authentication and active-window/device monitoring.	No single abstract-level classification number.	Demonstrates that typing and device behavior can be integrated with visual/audio proctoring, but not

(c) Person/object detection for phones, notes, extra people, or suspicious aids

Paper	Method and data	Reported results or limitations	Relevance
Automated Online Exam Proctoring — Atoum et al., 2017 ¹	Text detection and phone detection, particularly using the wearcam field of view; also user verification and active-window monitoring.	OEP dataset: 24 subjects/sessions and five cheating categories.	Direct phone/text/environment detection, but the second camera is wearable POV and no mouse trajectories.
A Novel Deep Learning-Based Online Proctoring System Using Face Recognition, Eye Blinking, and Object Detection Techniques — Ahmad et al., 2021 ¹⁴	HOG face detector, OpenCV face recognition, eye-blink detection for stationary-picture attacks, and object detection for mobile phones, laptops, iPads, and books.	Uses Fddb and LFW; reports up to 97% face-detection accuracy and 99.3% face-recognition accuracy.	Online face/object proctoring; no mouse channel or separate environment camera in the abstract.
An Automated Online Proctoring System Using Attentive-Net — Potluri et al., 2023 ¹³	Multiple-person detection, liveness, head pose, and visual detection of mobile/book/laptop cases; CIPL and custom datasets.	Combined accuracy reported as 0.87.	Person and device context in online video, but not an independent environment camera or interaction telemetry.
Automated Online Proctoring System Using Gaze View Tracking and Custom Object Detection — Li and Li, 2022 ⁴	Two-camera gaze-view tracking plus configurable TensorFlow custom object detector for cheating targets.	Prototype trials described as effective; exact numeric metrics unavailable in the accessible abstract.	Strong gaze-plus-object precedent with a second camera; no mouse/keystroke.
Enhancing Academic Integrity in Online Assessments: Introducing an	Camera and microphone data; face orientation, spoofing, mouth tracking, audio	Procedia Computer Science, 235, 1399–1408. The accessible	Online environment/object/person detection; no mouse/keystroke

Effective Online Exam Proctoring Model Using YOLO — Singh, Nair, Babu, and Duraisamy, 2024 22	analysis, mobile-device recognition, and person counting. Pre-trained YOLO weights support object/person detection; exams terminate when a device, multiple persons, or no person is detected for 10 seconds.	abstract does not state a single final accuracy.	channel and no explicitly separate room camera.
Multi-Perspective Adaptive Paperless Examination Cheating Detection System — Hu et al., 2024 5	Lightweight-YOLOv5-CA detects books, paper, phones, and people from overhead and horizontal views; gaze model and BP/MLP fuse the evidence.	95% average recognition and 35 FPS in online simulation.	Best evidence that a distinct environment-camera arrangement can be fused with gaze, but no mouse telemetry.
Multimodal Transformer Framework for Real-Time Cheating Detection — Alkinani, 2026 7	Video Swin-Transformer and Audio Spectrogram Transformer with cross-modal temporal fusion; auxiliary cues include gaze, speech, and phone-related evidence.	Macro-F1 0.94, AUROC 0.97, 15–20% false-positive reduction, at least 25 FPS GPU.	Advanced multimodal object/phone and gaze-related reasoning, but no mouse/keystroke stream is established.

(d) CCTV or physical examination-hall surveillance

Paper	Method and data	Reported results or limitations	Relevance
Automation of Traditional Exam Invigilation Using CCTV and Bio-Metric — Hoque et al., 2020 19	Biometric reader before hall entry; 360-degree CCTV cameras, high-sensitivity microphones, and speakers during the exam; remote invigilator supervises multiple halls.	The abstract presents a framework and does not report a machine-learning accuracy.	Direct CCTV/physical-hall deployment precedent, but not automated gaze/mouse fusion.

Implementation of an Intelligent Exam Supervision System Using Deep Learning Algorithms — Mahmood et al., 2022 18	Faster R-CNN/Inception-V2 for suspicious activity based on head movement; MTCNN for face detection/recognition. Self-generated 5,000-image invigilation dataset.	99.5% training accuracy and 98.5% testing accuracy; 12 misclassified images out of 1,000; 94% reported at 150 students versus 98.5% at 100 or fewer.	Strong physical-hall computer-vision baseline; no mouse/gaze estimator or explicit online-to-CCTV extension.
Detection of Anomalous Behavior in an Examination Hall Towards Automated Proctoring — Soman, Devi, and Srinivasa, 2017 20	Indexed record describes surveillance-camera video analysis for anomalous behavior in an examination hall.	IEEE conference record; exact model and performance should be verified from the publisher PDF before relying on numeric claims.	Early direct CCTV/automated-hall precedent; no online multi-channel fusion established.
Development of an Examination Hall Malpractice Detection System Using Pose Estimation and Machine Learning — Noma-Osaghae et al., 2025 21	OpenPose human-pose estimation fused with YOLO object detection for body movement, gestures, unauthorized devices, and written materials; integrated into a web application.	Accuracy 91.71%, precision 90.74%, F1 94.07%, recall 97.64% on a curated diverse examination-setting dataset.	Strong pose-plus-object physical-hall precedent; no gaze/interaction telemetry or explicit CCTV-to-online extension.
Cheating Recognition in Examination Halls Based on Improved YOLOv8 — Xu, Lu, Xu, Wang, et al., 2025 23	Surveillance-video keyframe selection via frame differencing and adaptive CNN thresholds; improved YOLOv8 with MLP and SENetV2; ResNet/3-D convolution behavior recognition and pose estimation. Custom CAN dataset plus real hall footage.	Article text reports approximately 45 FPS and above 90% accuracy; keyframe module reports 89.7% suspicious-behavior recall and more than 99% critical-event capture at 3–5% frame retention.	Direct recent physical surveillance prior art; no mouse/keystroke and no dedicated gaze channel.

Multi-Perspective Adaptive Paperless Examination Cheating Detection System — Hu et al., 2024 5	Three distinct viewpoints: overhead, face, and horizontal; object and gaze fusion. Although framed as paperless/online simulation, its multi-perspective physical arrangement is relevant to examination-hall cameras.	35 FPS and 95% average recognition in simulation.	Bridges online and physical viewpoints more than most sources, but no explicit CCTV deployment proposal and no mouse channel.
Automation of Traditional Exam Invigilation Using CCTV and Bio-Metric — Hoque et al., 2020 19	360-degree CCTV plus biometric hall entry and audio communication.	Conceptual/architectural abstract; no reported ML result.	Provides the clearest direct precedent for a CCTV examination-center operational model, but not for AI fusion with gaze and mouse.
Cheating Recognition in Examination Halls Based on Improved YOLOv8 — Xu et al., 2025 23	Surveillance video, keyframe mining, object detection, pose behavior recognition.	Custom CAN dataset and real-life hall video; final metrics require careful reading of all result tables because the abstract emphasizes architecture more than a single headline number.	Supports the physical-hall component and emphasizes robustness to occlusion, lighting, and subtle gestures.

Comparison table — most relevant 10–15 papers

The table below uses **Y**, **N**, and **Unclear** conservatively. “Mouse” is marked Y only when the accessible record explicitly describes mouse movement or a mouse/interaction stream. Active-window or tab-switch data alone are not treated as mouse dynamics.

Paper	Year	Gaze	Mouse / interaction	Environment camera
Atoum et al., <i>Automated Online</i>	2017	Y	Active-window, not confirmed mouse	Phone/text v wearcam

Exam Proctoring 1			dynamics	
Abozaid and Atia, <i>Multi-Modal Online Exam Cheating Detection</i> 2	2022	Y	Unclear ; mouse events mentioned in secondary records	Object detec
Kaddoura and Gumaiei, <i>Towards Effective and Efficient Online Exam Systems</i> 3	2022	Not established	N	Back camera
Li and Li, <i>Gaze View Tracking and Custom Object Detection</i> 4	2022	Y	N	Follower camera/obje
Hu et al., <i>Multi-Perspective Adaptive Paperless Examination</i> 5	2024	Y	N	Overhead an horizontal
Li et al., <i>Visual Analytics Approach</i> 6	2021	Head movement, not gaze estimator	Y	Video record
Labayen et al., <i>Multimodal Biometrics Technology</i> 9	2021	N/unclear	Typing and active-window/device data	Webcam monitoring
Ferdosi et al., <i>Behavioral Patterns of Students</i> 10	2023	Y	N	Webcam/faci
Yaqub et al., <i>Proctoring Online Exam Using Eye Tracking</i> 11	2023	Y	N	Webcam, pri masked face
Dilini et al., <i>Browser-Based Online Exams</i>	2021	Y	Browser plug-in, not mouse	Built-in webc only

<i>Through Eye Gaze Tracking</i> 12				
Potluri et al., <i>Attentive-Net</i> 13	2023	Head pose/attention proxy	N	Live examine video
Ahmad et al., <i>Face Recognition, Eye Blinking, and Object Detection</i> 14	2021	Blinking/face cues, not dedicated gaze	N	Webcam/object context
Hosny et al., <i>Mouse Interaction Behavior</i> 15	2022	N	Y	None
Mahmood et al., <i>Intelligent Exam Supervision</i> 18	2022	Head movement	N	Classroom/surveillance images
Hoque et al., <i>CCTV and Bio-Metric</i> 19	2020	N	N	360-degree C
Alkinani, <i>ViProctor-Transformer</i> 7	2026	Auxiliary gaze cues	Behavioral claim, but mouse not established	Wearable car
Taj et al., <i>Three-Eyed Invigilator</i> 8	2026	Not established in abstract	N	Front/rear m cameras and rotator
Xu et al., <i>Improved YOLOv8 Examination Halls</i> 23	2025	Pose, not dedicated gaze	N	Surveillance camera

Novelty verdict and recommendations

1. Is there an existing paper that does all three channels together?

No verified instance was located in this search. The conclusion is deliberately qualified because literature searches cannot prove universal nonexistence, and because some publisher pages—especially IEEE Xplore—were access-blocked. The reviewed evidence instead reveals several near-neighbors:

Atoum et al. [1](#) combine gaze, active-window monitoring, phone/text detection, audio, and two cameras. This is the closest historical integrated architecture, but active-window monitoring is not demonstrated as continuous mouse dynamics, and the wearcam is a candidate-POV device rather than a fixed, separately angled environment/CCTV camera.

Hu et al. [5](#) combine gaze with three distinct viewpoints and object/person detection. This is the closest verified precedent for a face camera plus independent environment views, but it has no mouse or keystroke stream.

Li et al. [6](#) combine exam video and mouse movement analytics. This is the closest direct mouse-plus-video precedent, but it does not establish gaze estimation or a separate environment camera.

Abozaid and Atia [2](#) deserve a careful full-text check because secondary records mention mouse events, while the accessible indexed abstract clearly lists head pose, object detection, and eye-gaze estimation. Until the publisher paper is inspected, it should be described as **potentially relevant but not verified as all-three prior art**.

2. Is there a paper proposing the environment camera as CCTV-extensible for physical exam centers?

No clear, verified paper in the reviewed set makes that exact cross-context proposal.

Several papers establish the two ends separately. Hoque et al. [19](#) propose a physical examination-hall system built around 360-degree CCTV, biometric entry, microphones, and speakers. Mahmood et al. [18](#), Noma-Osaghae et al. [21](#), and Xu et al. [23](#) develop computer-vision surveillance for physical halls. Atoum et al. [1](#), Hu et al. [5](#), Li and Li [4](#), and Taj et al. [8](#) develop multiple-camera or environmental-view systems for online or simulated remote assessment.

The missing claim is not “CCTV exists in exam halls” —it clearly does—but rather that **the same synchronized three-channel architecture is designed to operate first in remote online exams with a distinct environment camera and then in physical or hybrid exam centers by replacing that camera with a fixed CCTV feed**. That is a plausible deployment contribution, but it should be supported by a formal architecture, camera-calibration protocol, threat model, and cross-domain experiment.

3. Classification of the proposed contribution

The most accurate classification is:

Novel in combination and deployment architecture, not novel in individual components.

The individual components are established as follows:

Proposed component	Prior-art status	Evidence
Webcam gaze/head/eye analysis	Established	Atoum et al. 1 , Dilini et al. 12 , Yaqub et al. 11 , Ferdosi et al. 10 , Hu et al. 5
Mouse/interaction behavior	Established	Li et al. 6 , Hosny et al. 15 , Diedenhofen and Musch 17
Keystroke/typing behavior	Established	Morales and Fierrez 16 , Labayen et al. 9
Phone/book/note/person detection	Established	Atoum et al. 1 , Ahmad et al. 14 , Li and Li 4 , Hu et al. 5 , Singh et al. 22
Separate second/third camera	Established	Atoum et al. 1 , Kaddoura and Gumaï 3 , Li and Li 4 , Hu et al. 5 , Taj et al. 8
CCTV physical-hall surveillance	Established independently	Hoque et al. 19 , Mahmood et al. 18 , Noma-Osaghae et al. 21 , Xu et al. 23
Exact gaze + mouse + independent environment-camera fusion	No verified instance located	Gap indicated by the comparison above; Abozaid and Atia 2 requires full-text confirmation
Explicit online-to-CCTV substitution architecture	No clear verified instance located	Existing online and physical systems are mostly separate 1 5 18 19

4. Recommended sharper novelty claims

Recommendation 1: Claim a defined three-stream fusion model, not merely “multimodal AI.” Specify exactly how the streams are synchronized and fused. For

example, each time window could contain gaze deviation duration and dispersion, head-pose consistency, cursor velocity/acceleration, click and idle statistics, tab/window-focus events, and environment-camera detections. Compare early fusion, late probability fusion, and temporal cross-attention or gated fusion. The novelty then resides in the *fusion formulation and evidence calibration*, rather than in owning the sensors.

Recommendation 2: Make the third camera an explicit hardware and geometry contribution. Define a reproducible placement, such as a front-facing webcam for face/gaze and a side or rear three-quarter camera covering the desk and doorway. Report the field of view, camera height, angle, synchronization mechanism, resolution, frame rate, and blind-zone analysis. Compare one-camera, two-camera, and three-camera ablations. Hu et al. [5](#) demonstrates why this matters: camera viewpoint determines which areas of the workspace are visible.

Recommendation 3: Introduce an explainable, risk-calibrated alert score. Do not treat one glance, one tab switch, or one detected phone as definitive cheating. Use a temporal score with persistence, modality reliability weights, uncertainty estimates, and human-review thresholds. Report false-positive rate by benign behavior—reading, looking at scratch paper, adjusting posture, accessibility aids, poor lighting, glasses, or network delay—and use calibrated precision-recall, AUROC, macro-F1, and event-level detection metrics.

Recommendation 4: Build a new synchronized dataset with realistic benign and malicious behavior. The MSU OEP dataset is valuable but small and based partly on enacted behavior [1](#) [7](#). A contribution could include synchronized face-camera, environment-camera, screen/window telemetry, cursor events, and ground-truth event intervals. Include real and simulated exams, multiple room layouts, different camera placements, glasses, lighting, accessibility devices, and varied computer hardware. Release annotation guidelines and participant-consent/privacy procedures.

Recommendation 5: Treat online-to-physical deployment as domain adaptation. A remote environment camera and a fixed hall CCTV stream have different perspectives, numbers of candidates, illumination, occlusion patterns, and identity-association problems. Define the deployment as a modular camera-adaptor layer and test whether a model trained on single-candidate remote sessions transfers to multi-candidate hall footage. If no experiment is available, present CCTV use as a future extension rather than as an experimentally demonstrated contribution.

5. Limitations of this assessment

This is a literature-based novelty assessment, not a patentability opinion. The search prioritized publisher pages, IEEE/ACM records, Springer, ScienceDirect, PubMed-indexed articles, institutional repositories, and open-access papers. Some publisher pages were blocked by automated traffic controls or paywalls; in those cases, conclusions were limited

to accessible abstracts, repository records, or authoritative project pages. Abozaid and Atia ² is the main unresolved near-neighbor because its indexed abstract confirms gaze/head-pose/object modalities while other records mention mouse events. AutoOEP is included only as a cautionary recent lead: its arXiv record is explicitly marked withdrawn and should not be treated as confirmed peer-reviewed evidence ²⁵.

The phrase “no verified paper located” should therefore be used in the manuscript instead of “no one has ever proposed this.” A systematic review should also document database names, exact queries, dates, duplicate removal, inclusion/exclusion rules, and full-text screening results so that the novelty claim is reproducible.

Plain reference list — IEEE style

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